

①

TOPIC

DATE

3-9-23

Chemistry

Section - A

1. Argon
2. Calcium
3. 2.2

$$\text{Molecular wt} = 2 \times V.D$$

$$12 + 2 \times 16 = 2 \times V.D$$

$$\frac{12 + 32}{2} = V.D$$

2

$$\frac{22}{2} = V.D$$

4. 17

5. 1:2

6. Mercury

7.

8. 1

9. Lithium

10.

11. Copper chloride

12. Consists of molecules

13. An acid

14.

15. $\text{HNO}_3 + \text{H}_2\text{SO}_4$

16. i) a) Y

b) L

c) Halogens

d) Boron - group 13

Nitrogen - group 15

Teacher's Signature.....

(2)

TOPIC

DATE

e) Oxygen (O)

ii) a) reducing agents

b)

c) acidic

d) silver chloride

e) Cl

$$\begin{aligned}
 \text{iii) i) Mol. wt} &= 32 + 2 \times 16 \\
 &= 32 + 32 \\
 &= 64 \text{ g}
 \end{aligned}$$

64 g of SO_2 = 1 mol of SO_2

$$\begin{array}{ccc}
 \therefore 64 \text{ g} & \text{VOLUME} & \\
 320 \text{ g} & \begin{array}{c} 22.4 \text{ l} \\ \nearrow \quad \searrow \\ x \end{array} &
 \end{array}$$

$$x = \frac{11.2 \times 10}{22.4 \times 320}$$

$x = 112 \text{ l}$ ^{the volume} is occupied by SO_2

ii) Gay-Lussac's law states that when gases react they do so in volumes that bears a simple whole number ratio with temperature and pressure remaining constant.

$$\begin{aligned}
 \text{iii) Mol. wt of } C_3H_8 &= 12 \times 3 + 1 \times 8 \\
 &= 36 + 8 \\
 &= 44 \text{ g}
 \end{aligned}$$

32 g of O_2 occupies 22.4 l volume

$\therefore$ 1 mole of O_2 = 32 g

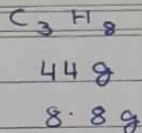
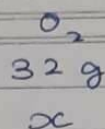
Teacher's Signature.....

(3)

TOPIC

DATE

0-2



$$\begin{array}{r} x = \frac{2 \times 32}{8 \times 32} \\ \frac{44}{11} \end{array}$$

$$= 6.4 \text{ g of O}_2$$

is needed for complete combustion.

iv) i)

ii)



$$64 + 32 + 4 \times 16 + 5 \times 18$$

$$96 + 64 + 90$$

$$= 250 \text{ g}$$

$$\begin{array}{r} 96 \\ 64 \\ \hline 90 \\ 250 \end{array}$$

$$\therefore \% \text{ of H}_2\text{O} = \frac{90}{250} \times 100 = 36 \%$$

iii)

Molecular

$$\text{formula} = [\text{X}_x\text{Y}_y]_n$$

$$\text{V.D} = \frac{\text{Empirical formula weight}}{\text{Molecular weight}}$$

$$n =$$

$$\frac{\text{Molecular weight}}{\text{E.F}}$$

$$\frac{2 \times \text{V.D}}{\text{V.D}}$$

$$\therefore n = 2$$

$$\therefore \text{Mol. formula} = (\text{X}_x\text{Y}_y)_2 = \text{X}_2\text{Y}_4$$

v. i)

15

ii)

19

iii)

8

iv)

4

v)

2

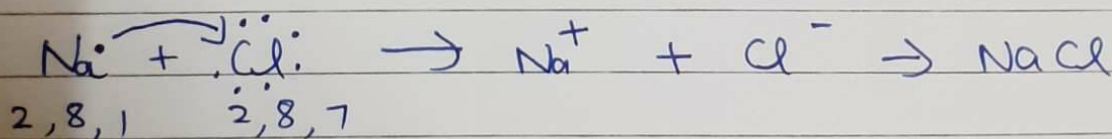
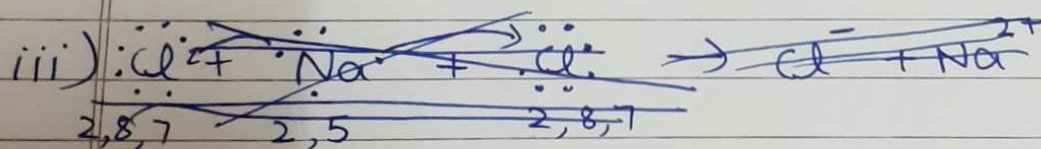
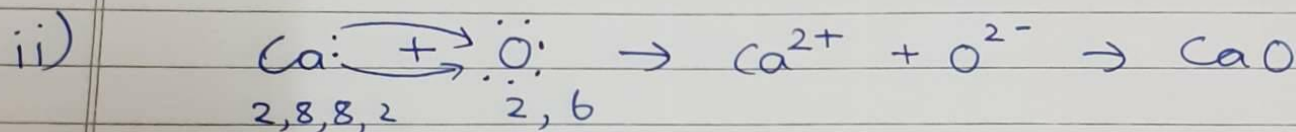
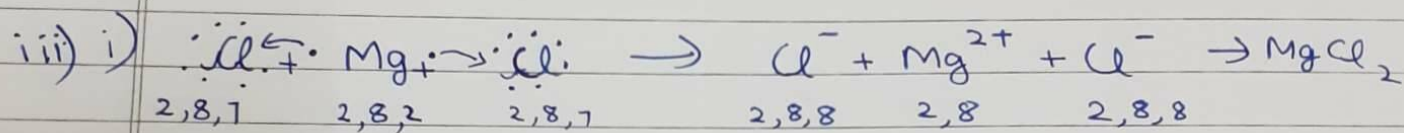
Teacher's Signature.....

Section - B

3. i) reaction with sulphur $\rightarrow$ MS
 reaction with Chlorine $\rightarrow$ MCl₂

ii) a) Molecules of hydrogen and oxygen have single covalent bonds between their atoms because they share only one pair of electrons between them whereas oxygen shares 2 pairs of electrons between its atoms.

b) A molecule of CCl₄ has 4 single bonds as the carbon atoms share ~~one~~ ^{one pair of} electrons with four chlorine atoms.



- iv. A co-ordinate bond is a type of covalent bond in which the shared pair of electrons comes from the same element.
 A coordinate bond is formed only when a neutral atom accepts the lone pair of electrons of another element. Eg: Hydronium ion.

Teacher's Signature.....

4. i) a) 2, 8, 8, 2
b) group 2, period 4

- ii) a) valency = 2
b) ionic compound

iii) Under same conditions of temperature and pressure equal volumes of all gases contain same number of molecules.

Applications of Avogadro's law:

- * Determines the atomicity of the gas
- * Shows the relation between molecular weight and vapour density of a gas.
- * Explains Gay Lussac's law.

iv. i) A and B

ii) A and C

iii) B and D

2. i) 1. True, as the cations lose their electrons to attain a stable configuration.

2. ~~True~~ True, because these form an electrovalent or ionic bond.

ii) 1 mole of CO_2 = gram mol. wt of CO_2

$$\therefore 1 \text{ mole of } \text{CO}_2 = 12 + 2 \times 16 \\ = 12 + 32$$

$$1 \text{ mole of } \text{CO}_2 = 44 \text{ g}$$

$$x = 22 \text{ g}$$

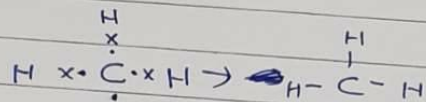
$$x = \frac{22 \times 44}{44} = 22 \text{ g}$$

Teacher's Signature.....

= 0.5 mole

- iii) a) Oxidising power of X is greater than Y.
 b) Electronegativity of X is greater than Y.
 c) X is likely to be placed to the right of Y in the periodic table.

iv. i) CH_3



Carbon ~~Methane~~ forms a single covalent bond with each hydrogen atom to form one molecule of CH_4 .

- ii) a) Be will form ions readily as it is easy to lose electrons from its outermost shell.

b) All of these elements have the same number of electrons in their outermost shell that is 2 electrons in the valence shell. So they have a valency of 2 and belong to group two of the periodic table.

6. i) a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

b) Anhydrous CuCl_2

ii. Deliquescent ~~int~~ salts ^{absorb} ~~lose~~ their water from the atmosphere and dissolve in it, this is called deliquescence.

Efflorescent salts lose their water of crystallisation to the atmosphere completely and turn into a powder form.

- iii. * Acids neutralize bases to give salt and water
 * Acids react with metals to ~~for~~ liberate hydrogen gas.

Teacher's Signature.....

TOPIC

DATE

* ~~More~~ ^{less} volatile acids replace ~~less~~ ^{more} volatile acids on heating with salts.

- iv. a) ~~Tribasic~~ Tribasic acid
b) diacidic ~~base~~ base
c) monoacidic base
d) monobasic acid
e) dibasic acid
f) Tribasic ~~an~~ acid.